

Supplementary Material

Estimating Infectious Disease Importation Risk during the 2026 FIFA World Cup in the United States, June–July 2026

Appendix A: Detailed Methods and Data Sources

Study setting

The 2026 FIFA World Cup will be hosted jointly by the United States, Mexico, and Canada, with matches scheduled from June 11 through July 19, 2026. The United States will host the majority of matches across eleven venues: MetLife Stadium (East Rutherford, NJ), SoFi Stadium (Inglewood, CA), AT&T Stadium (Arlington, TX), Mercedes-Benz Stadium (Atlanta, GA), NRG Stadium (Houston, TX), Arrowhead Stadium (Kansas City, MO), Lincoln Financial Field (Philadelphia, PA), Hard Rock Stadium (Miami Gardens, FL), Levi’s Stadium (Santa Clara, CA), Lumen Field (Seattle, WA), and Gillette Stadium (Foxborough, MA). Canada hosts two venues (BC Place, Vancouver; BMO Field, Toronto) and Mexico hosts three (Estadio Azteca, Mexico City; Estadio BBVA, Monterrey; Estadio Akron, Guadalajara). The geographic distribution of all 16 venues across the three co-host nations is shown in Figure 1. All 48 qualified nations are listed in Supplementary Table 1.

We estimated the importation risk of five infectious diseases—dengue fever, influenza, malaria, measles, and pertussis—for each host city using three complementary models of increasing resolution: (1) a *baseline model* using COR June 2024 arrivals with BTS T-100 routing fractions and no World Cup growth adjustment; (2) a *World Cup (WC)-adjusted model* that scales volumes to country-level 2026 projections; and (3) a *schedule-driven model* that separates World Cup fan travel from background tourism and routes fans to the specific venue cities where their national team plays. The five diseases were selected because each is endemic in countries with strong football traditions, each has a plausible pathway for travel-associated importation into the United States, and each represents a distinct surveillance and public health response challenge. Influenza is included because June corresponds to the

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16 venues across USA (11), Canada (2), and Mexico (3)

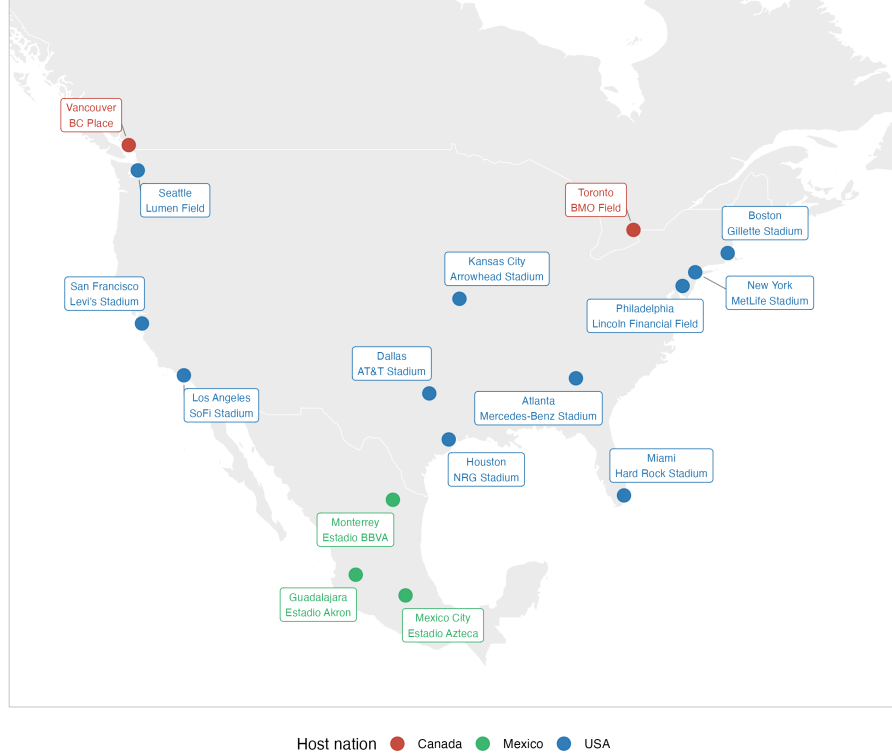


Figure 1: Geographic distribution of the 16 FIFA World Cup 2026 venue stadiums across the three co-host nations. Circles are colour-coded by host country: USA (blue, 11 venues), Canada (red, 2 venues), and Mexico (green, 3 venues). Suburb stadium locations are labelled by their metropolitan area to match the spatial aggregation used throughout the analysis (e.g. East Rutherford → New York, Foxborough → Boston).

Southern Hemisphere winter peak, coinciding with the tournament window for WC-qualified nations in South America and Oceania.

Notation and variable definitions

The following indices and variables are used consistently throughout all three models. Where the range of an index differs between models, this is noted explicitly.

Indices

- c — source **country** of international travelers (all three models).
- v — **destination venue city**. In all three models, v ranges over the eleven US WC venue cities (Table 1). Model 3 additionally covers the five non-US host cities (Toronto, Vancouver, Guadalajara, Mexico City, Monterrey).

- d — **disease**: $d \in \{\text{dengue, influenza, malaria, measles, pertussis}\}$.
- y — **calendar year**, used when averaging over multiple June cross-sections ($y \in \{2023, 2024, 2025\}$ for T-100 routing fractions).

Travel and importation variables

- $N_{c,v}$ — expected number of **travelers** from source country c arriving in city v during June. Constructed differently in each model (see below).
- $\Lambda_{v,d}$ — **importation intensity** (Poisson rate): expected number of imported cases of disease d arriving in city v during June 2026.
- $\lambda_{c,v,d}$ — **importation contribution** from source country c to city v for disease d ; $\Lambda_{v,d} = \sum_c \lambda_{c,v,d}$.

Disease and parameter variables

- $I_{c,d}$ — **per-traveler infection probability**: probability that a randomly selected traveler from country c is carrying disease d during their trip. Defined as $I_{c,d} = \rho_d \cdot C_{c,d}/P_c$.
- $C_{c,d}$ — **reported cases** of disease d in source country c during June (or annual figure divided by 12 for diseases with only annual reporting).
- P_c — **population** of source country c .
- ρ_d — **reporting adjustment factor** for disease d : corrects for under-reporting in routine surveillance and the fraction of reported cases that are actively infectious at any given time (Table 2).
- p_d — **travel-while-infectious probability**: probability that a person infected with disease d continues to travel internationally during the infectious window (Table 2).

Routing fractions and travel growth

- $f_{c,v}^{\text{T100}}$ — **T-100 routing fraction**: share of country c 's US-bound June arrivals that enter through city v , derived from BTS T-100 International Segment data averaged over June 2023–2025. Used in all three models.
- ϕ_c — **growth factor** for country c : ratio of projected 2026 to observed 2024 annual visitor volume ($\phi_c = V_{c,2026}/V_{c,2024}$). Derived from NTTO for Tier 1 countries; set to ϕ_{WC} or ϕ_{base} for all others.

- $\phi_{\text{WC}} = 1.174$ — aggregate WC growth factor (NTTO total: 85,017/72,390 thousand).
- $\phi_{\text{base}} = 1.077$ — baseline (non-WC) growth factor derived as the geometric mean annual growth rate over June 2023–2025, averaged across COR (1.062) and T-100 (1.092) data sources; applied to non-qualified countries.
- N_c^{WC} — **WC-fan travel increment**: marginal arrivals attributed to the World Cup for country c .
- N_c^{bg} — **background arrivals**: June 2026 travel that would have occurred without the WC.
- $\omega_{c,v}$ — **schedule routing fraction**: share of WC fans from country c attending matches in venue city v ; defined as games played in v divided by total known games for that team.
- $\mathcal{Q}$ — set of WC-qualified countries with a known group-stage match schedule (TBD matches excluded).

Importation risk framework

We modelled disease importation as a Poisson process. This choice is standard for rare-event counting problems: when individual import events are independent and the per-traveller probability of infection is small, the number of imported cases in a fixed time window converges to a Poisson random variable. The Poisson framework is also analytically tractable and additive over source units, allowing country-level and stream-level contributions to be computed and summed independently.

If $\Lambda_{v,d}$ is the expected number of imported cases of disease d arriving in city v during June 2026, the probability of observing at least one importation is:

$$P(X_{v,d} \geq 1) = 1 - e^{-\Lambda_{v,d}} \quad (1)$$

where $\Lambda_{v,d} = \sum_c \lambda_{c,v,d}$ sums over all origin countries c . This framework is consistent with previously published importation risk analyses for large international sporting events [1].

Each country-level contribution takes the same multiplicative form across all three models:

$$\lambda_{c,v,d} = N_{c,v} \cdot I_{c,d} \cdot p_d \quad (2)$$

where $N_{c,v}$ is the number of travellers from country c arriving in city v , $I_{c,d}$ is the per-traveller probability that an arriving person carries disease d , and p_d is the probability that such a

91 person travels internationally while infectious. The three models differ in how $N_{c,v}$ and $I_{c,d}$
 92 are constructed.

93 **Baseline model: COR June 2024 travel volume with T-100 routing** 94 **(no WC adjustment)**

95 **Rationale and design**

96 The baseline model establishes the importation risk expected in a typical June *without* any
 97 World Cup effect. Its purpose is to provide a pre-tournament reference against which the
 98 WC-driven changes in Models 2 and 3 can be measured. Three design choices ensure this
 99 reference is both accurate and spatially consistent:

- 100 1. **Travel volume: COR June 2024, no growth factor.** Monthly air arrivals by
 101 country of residence (COR/I-94) for June 2024 are used directly, without any WC
 102 growth factor ($\phi_c = 1$ for all countries). June 2024 is the most recent pre-tournament
 103 June with complete data and represents normal summer travel patterns.
- 104 2. **City routing: BTS T-100 routing fractions.** Country-level routing fractions
 105 $f_{c,v}^{\text{T100}}$ (described in the next subsection and in Equation 7) allocate each country's
 106 COR arrivals to the eleven US venue cities. Using T-100 from the baseline ensures
 107 all three models operate on the same spatial grid, making the three-way comparison
 108 interpretable: the step from Model 1 to Model 2 isolates the WC travel surge alone,
 109 not a change in routing methodology.
- 110 3. **Disease incidence: country-level.** Country-specific incidence proxies $I_{c,d}$ are used
 111 (Equation 11), providing the same precision as Models 2 and 3 and ensuring that
 112 differences between models reflect travel volume changes, not incidence aggregation.

113 **Baseline city-level arrivals**

114 Baseline arrivals from country c to venue city v are:

$$N_{c,v}^{\text{baseline}} = N_{c,\text{June 2024}}^{\text{COR}} \cdot f_{c,v}^{\text{T100}} \quad (3)$$

115 Countries with no T-100 routing data (typically nations with no scheduled nonstop in-
 116 ternational service to the United States) are excluded; their volumes are negligible in the
 117 aggregate.

Importation intensity (baseline)

$$\lambda_{c,v,d} = N_{c,v}^{\text{baseline}} \cdot I_{c,d} \cdot p_d \quad (4)$$

$$\Lambda_{v,d} = \sum_c \lambda_{c,v,d} \quad (5)$$

The probability of at least one importation is computed via Equation 1.

BTS T-100 International Segment data: country-level city routing fractions

Motivation and data source

All three models require country-level city routing fractions that cover every US WC venue city. We use the Bureau of Transportation Statistics (BTS) T-100 International Segment (All Carriers) dataset [2], which records nonstop international flight segments landing at US airports at the individual airport level. T-100 covers all US commercial airports and reports passengers by country of origin (via the departure country code), making it possible to compute country-specific routing fractions for all eleven US venue cities. The T-100 data do not include an origin city field, only the originating country code for the international segment.

Data processing and routing fraction construction

Three full years of T-100 data were downloaded for 2023, 2024, and 2025, covering all months. Only June records (month = 6) were retained for each year, yielding three June cross-sections. This window was chosen to: (1) match the June travel period of the WC group stage; (2) use the three most recent complete pre-tournament June months; and (3) avoid the COVID-recovery period (2020–2022) during which route networks and traffic volumes were atypical.

Records were filtered to scheduled passenger service only (`CLASS = "F"`), excluding charter (`N`), all-cargo (`L`), and mixed cargo (`P`) service classes. Only segments with at least one passenger (`PASSENGERS > 0`) and landing in the United States (`DEST_COUNTRY = "US"`) were retained. Destination airports were then mapped to the eleven US WC venue city groups using a predefined airport-to-city table (e.g., JFK, EWR, and LGA → New York; MIA and FLL → Miami; SFO, OAK, and SJC → San Francisco). Any airport not associated with a WC venue city was excluded.

For each source country c , venue city v , and year y , the routing fraction is first computed within each June cross-section as:

$$f_{c,v,y}^{\text{T100}} = \frac{N_{c,h,y}^{\text{June}}}{N_{c,\text{US},y}^{\text{June}}} \quad (6)$$

where $N_{c,h,y}^{\text{June}}$ is the total T-100 passengers from country c to the airport group serving venue city v in June of year y , and $N_{c,\text{US},y}^{\text{June}}$ is the total T-100 passengers from country c to *all* US airports in June of year y (the denominator). Using the full US total—not just WC venue airports—ensures that fractions are true proportions, interpretable as the share of US-bound travel from country c that enters through each city.

The mean routing fraction is then averaged across the three years:

$$f_{c,v}^{\text{T100}} = \frac{1}{|Y|} \sum_{y \in Y} f_{c,v,y}^{\text{T100}} \quad (7)$$

where $Y = \{2023, 2024, 2025\}$. Averaging within years before averaging across years (rather than pooling all passenger counts) prevents high-volume years from dominating the estimate and keeps each annual fraction a valid proportion.

Important limitations of T-100 routing fractions

T-100 records the *nonstop international segment* only. A passenger travelling on a connecting itinerary (e.g., São Paulo → Miami → Kansas City) appears only in the Miami row, because Miami is the airport where the international segment terminates and US Customs is cleared. Kansas City, as a domestic connecting leg, is not recorded in T-100 for that passenger. This is the correct interpretation for importation risk modelling: from a public health perspective, the traveller enters the US in Miami, where surveillance and border health screening occurs. Kansas City correctly shows near-zero T-100 routing fractions for most countries, reflecting minimal direct international service; its WC-specific traffic is captured entirely by the schedule-driven WC-fan stream.

The T-100 data cover US airports only. Routing fractions for the five non-US host cities (Toronto, Vancouver, Guadalajara, Mexico City, Monterrey) cannot be derived from T-100 and are not estimated here; those cities receive importation risk estimates solely from the WC-fan travel stream.

Country names in T-100 use ISO/ICAO conventions (e.g., “Korea, Republic of”, “Viet Nam”) while the CBP COR arrival data use informal short names (e.g., “South Korea”, “Vietnam”). A standardised recode table covering 26 known mismatches was applied before

173 joining the two datasets.

174 **World Cup–adjusted model: country-level travel volume (2026 pro-** 175 **jections)**

176 **Motivation**

177 The baseline model uses June 2024 COR arrivals without any World Cup adjustment, cap-
178 turing normal summer travel patterns. It does not account for the additional influx of
179 visitors attending the World Cup. The WC-adjusted model addresses this limitation by
180 projecting June 2026 arrival volumes at the country level using NTTD forecasts and COR
181 data scaled by WC growth factors. All other modelling choices (T-100 routing fractions,
182 country-level disease incidence, same ρ_d and p_d values) are unchanged, so the Model 1 to
183 Model 2 comparison isolates the effect of the WC travel surge alone.

184 **Construction of 2026 travel volume**

185 Country-level June 2026 visitor volumes to the United States were constructed from two
186 complementary sources using a tiered approach. The tiered design is necessary because not
187 all countries have official NTTD projections; for the majority of the world, the only available
188 country-level arrival data are the CBP COR monthly figures, which are then scaled by an
189 appropriate growth factor.

190 **Tier 1 — NTTD projections (top 12 source markets).** Official 2026 international
191 visitor projections for twelve major source markets were obtained from the US National
192 Travel and Tourism Office (NTTD) 2025 Forecast Report [3]. These projections—available
193 for Australia, Brazil, Canada, China, France, Germany, India, Italy, Japan, Mexico, South
194 Korea, and the United Kingdom—already incorporate the World Cup tourism effect and
195 should *not* have additional WC visitors added on top. The 2024-to-2026 growth factor
196 for each country was derived as $\phi_c = V_{c,2026}/V_{c,2024}$, where $V_{c,y}$ is the projected annual
197 visitor volume in year y (Table 3). Using the NTTD projections for these twelve countries
198 is important because they reflect expert forecasts that account for the specific WC effect,
199 bilateral travel trends, and macroeconomic factors, making them more reliable than a simple
200 scaling of 2024 COR arrivals.

201 **Tier 2 — COR data scaled by global growth factors (remaining countries).** For
202 all other countries, country-level June 2024 arrivals were extracted from the US Customs and
203 Border Protection (CBP) I-94 Monthly Arrivals by Country of Residence dataset [4]. The

COR data provide monthly arrival counts by country of residence and are the most granular publicly available source of international arrival data for the United States. A growth factor was then assigned based on World Cup qualification status:

- *WC-qualified countries not in Tier 1*: scaled by the overall NTTO 2024→2026 growth factor ($\phi_{WC} = 85,017/72,390 = 1.174$). This factor was derived from the aggregate NTTO total and incorporates the WC tourism uplift at the global level, providing a reasonable approximation for WC-qualified countries not individually projected by NTTO.
- *Non-qualified countries*: scaled by a data-driven baseline growth factor ϕ_{base} estimated from observed trends in US inbound travel independent of the tournament. We derived ϕ_{base} from two sources covering June 2023–2025: (i) CBP COR monthly arrivals and (ii) BTS T-100 total international passengers arriving in the US. For each source, the geometric mean annual growth rate over the two observed intervals is:

$$\bar{g} = \left(\frac{N_{\text{June},2025}}{N_{\text{June},2023}} \right)^{1/2} = \sqrt{\frac{N_{2024}}{N_{2023}} \cdot \frac{N_{2025}}{N_{2024}}} \quad (8)$$

and the two-year forward factor follows as $\phi_{base} = \bar{g}^2 = N_{\text{June},2025}/N_{\text{June},2023}$. The COR estimate is $\phi_{base}^{COR} = 1.062$ (June totals: 4,970K in 2023, 5,279K in 2025) and the T-100 estimate is $\phi_{base}^{T100} = 1.092$ (passengers: 10,700K in 2023, 11,686K in 2025). The two sources agree closely; their arithmetic mean, $\phi_{base} = 1.077$, is used in the model. This value is lower than a simple extrapolation of pre-2024 growth because June 2025 arrivals declined relative to June 2024 in both datasets, reflecting a plateau after the post-COVID travel rebound rather than continued acceleration.

The projected June 2026 arrivals for country c are thus:

$$N_{c,\text{June } 2026} = N_{c,\text{June } 2024}^{COR} \cdot \phi_c \quad (9)$$

Country-to-city allocation via T-100 routing fractions. City-level arrival volumes for country c and venue city v were estimated by applying the T-100 country-level routing fractions (Equation 7):

$$N_{c,v}^{2026} = N_{c,\text{June } 2026} \cdot f_{c,v}^{T100} \quad (10)$$

This uses the same T-100 routing fractions as the baseline model (Equation 3), but applied to the projected 2026 volumes. Countries with no T-100 routing data are excluded from the city-level allocation; their volumes are negligible.

Country-level disease incidence

Disease burden is computed at the country level across all three model tiers, providing fine-grained precision that reflects within-region heterogeneity in disease burden. For example, Brazil and Argentina differ substantially in dengue incidence despite belonging to the same broad regional grouping. Country-level incidence is defined as:

$$I_{c,d} = \rho_d \cdot \frac{C_{c,d}}{P_c} \quad (11)$$

where $C_{c,d}$ is the reported cases of disease d in country c and P_c is the national population (2020 estimates from the UN World Population Prospects [5]). The same ρ_d and p_d values are used in all three models to ensure that model comparisons reflect travel volume differences alone.

Importation intensity (WC-adjusted)

$$\lambda_{c,v,d} = N_{c,v}^{2026} \cdot I_{c,d} \cdot p_d \quad (12)$$

$$\Lambda_{v,d} = \sum_c \lambda_{c,v,d} \quad (13)$$

The probability of at least one importation is then computed via Equation 1.

Schedule-driven venue routing model

Motivation

Both the baseline and WC-adjusted models distribute all international arrivals across venue cities using T-100 routing fractions, which reflect habitual tourist flow patterns. This is appropriate for background tourism but does not capture the WC-specific fan travel logic: a supporter from Brazil attending a group-stage match in Houston is unlikely to distribute to Boston, Philadelphia, or Seattle in the same proportions as a regular tourist. The schedule-driven model captures this distinction by decomposing total travel into a WC-specific fan stream—routed by the match schedule—and a background stream routed by T-100 fractions.

Travel decomposition

For each source country c , total projected June 2026 arrivals $N_{c,\text{June 2026}}$ (Equation 9) are decomposed into two components:

- *WC-fan stream*: the marginal arrivals attributable to the World Cup, defined as the increment above the no-WC counterfactual (the volume that would have arrived with only baseline growth):

$$N_c^{\text{WC}} = N_{c, \text{June } 2024}^{\text{COR}} \cdot \max(0, \phi_c - \phi_{\text{base}}) \quad (14)$$

This formulation isolates the WC-specific increment: any growth above the baseline rate ($\phi_{\text{base}} = 1.077$) is attributed to the tournament.

- *Background stream*: arrivals that would have occurred regardless of the World Cup:

$$N_c^{\text{bg}} = N_{c, \text{June } 2026} - N_c^{\text{WC}} \quad (15)$$

For countries with NTTO growth factors below the baseline ($\phi_c < \phi_{\text{base}} = 1.077$), the WC increment is set to zero and all travel is treated as background. For a WC-qualified country assigned the global WC growth factor ($\phi_{\text{WC}} = 1.174$), the WC fan fraction is approximately 8.3% of projected arrivals: $(1.174 - 1.077)/1.174 \approx 0.083$.

Schedule-based WC fan routing

The 2026 FIFA World Cup match schedule was extracted from official FIFA event records for all 48 qualifying nations [6]. Matches with at least one undetermined opponent (labelled “TBD” in the schedule) are excluded from the fan routing calculation; both group-stage and knockout-round entries with TBD opponents are dropped. For each WC-qualified country c with team t_c , a schedule routing fraction is defined as:

$$\omega_{c,v} = \frac{g_{c,v}}{G_c} \quad (16)$$

where $g_{c,v}$ is the number of matches played by team t_c at venue city v and $G_c = \sum_v g_{c,v}$ is the total number of known matches for that team. This fraction represents the share of WC fans from country c expected to be present in venue city v , under the assumption that fan travel is proportional to the number of matches played there. WC-fan arrivals at each venue city are then:

$$N_{c,v}^{\text{WC}} = N_c^{\text{WC}} \cdot \omega_{c,v} \quad (17)$$

Background stream routing via T-100 fractions

Background travellers (Equation 15) are assumed to follow normal routing patterns, independent of the match schedule. We use the T-100 country-level routing fractions (Equation 7)

to distribute background arrivals across venue cities:

$$N_{c,v}^{\text{bg}} = N_c^{\text{bg}} \cdot f_{c,v}^{\text{T100}} \quad (18)$$

This is consistent with the routing approach used in the baseline and WC-adjusted models. Because T-100 covers all 11 US venue cities, background travelers contribute to importation risk across the full spatial grid. Countries with no direct T-100 service to a given city have a zero routing fraction for that city, correctly reflecting minimal direct international service there.

Note on multi-team countries. England and Scotland are distinct World Cup participants but share a single country-of-residence entry (“United Kingdom”) in the COR arrivals data. Both teams’ match venues are combined under this single entry, and the routing fractions $\omega_{\text{UK},v}$ are computed over the union of their games. Because the fractions sum to one, no double-counting of UK arrivals occurs.

Venue cities. The schedule-driven model covers all sixteen host venues across the three co-host nations (Table 1). For all eleven US venue cities, both the WC-fan and background streams contribute to importation risk: the WC-fan stream routes fans via the match schedule ($\omega_{c,v}$) and the background stream routes tourists via T-100 fractions ($f_{c,v}^{\text{T100}}$). For the five non-US host cities (Toronto and Vancouver in Canada; Guadalajara, Mexico City, and Monterrey in Mexico), T-100 fractions are not available and only the WC-fan stream contributes; background tourist flows to those cities are not estimated.

Importation intensity (schedule-driven)

The total Poisson importation intensity for venue city v is the sum of contributions from both travel streams across all source countries:

$$\Lambda_{v,d} = \underbrace{\sum_{c \in \mathcal{Q}} N_{c,v}^{\text{WC}} \cdot I_{c,d} \cdot p_d}_{\text{WC-fan stream}} + \underbrace{\sum_c N_{c,v}^{\text{bg}} \cdot I_{c,d} \cdot p_d}_{\text{background stream}} \quad (19)$$

where $\mathcal{Q}$ denotes the set of WC-qualified countries with a known group-stage schedule. The probability of at least one importation is computed via Equation 1.

Table 1: WC 2026 host venues and their canonical city names used in the schedule-driven model. Cities with T-100 background routing are marked with †.

Stadium	City (canonical)	Country	T-100 bg?
MetLife Stadium	New York†	USA	Yes
SoFi Stadium	Los Angeles†	USA	Yes
AT&T Stadium	Dallas†	USA	Yes
Mercedes-Benz Stadium	Atlanta†	USA	Yes
NRG Stadium	Houston†	USA	Yes
Arrowhead Stadium	Kansas City†	USA	Yes*
Lincoln Financial Field	Philadelphia†	USA	Yes
Hard Rock Stadium	Miami†	USA	Yes
Levi’s Stadium	San Francisco†	USA	Yes
Lumen Field	Seattle†	USA	Yes
Gillette Stadium	Boston†	USA	Yes
BC Place	Vancouver	Canada	No
BMO Field	Toronto	Canada	No
Estadio Azteca	Mexico City	Mexico	No
Estadio BBVA	Monterrey	Mexico	No
Estadio Akron	Guadalajara	Mexico	No

* Kansas City shows near-zero T-100 fractions for most countries, reflecting its minimal direct international air service; its WC-specific traffic is captured by the WC-fan stream.

Three-model comparison

To quantify the effect of each modelling assumption, results from the three models are compared side by side for each disease and destination city. The three models form a nested hierarchy:

1. **Baseline:** COR June 2024 arrivals (no WC adjustment), country-level disease incidence, all eleven US venue cities via T-100 routing fractions. Establishes the pre-tournament reference.
2. **WC-adjusted:** Country-level June 2026 projections (NTTO + COR scaling by ϕ_c), same country-level disease incidence, same T-100 routing, all eleven US venue cities.
3. **Schedule-driven:** Same 2026 projections decomposed into WC-fan and background streams; WC fans routed to all sixteen venue cities via the match schedule ($\omega_{c,v}$); background travelers routed via T-100 fractions to all eleven US cities.

The step from Model 1 to Model 2 isolates the effect of the WC travel surge (growth factors ϕ_c applied to 2024 base volumes). All other modelling choices—routing fractions,

disease incidence, ρ_d , p_d —are held fixed. The step from Model 2 to Model 3 isolates the effect of schedule-based venue routing: WC fans are no longer spread by habitual tourist flows but directed to the specific cities where their team plays. The five non-US host cities (Toronto, Vancouver, Guadalajara, Mexico City, Monterrey) appear only in Model 3.

Results across all three model tiers are compared in Figure 5 (the dodged-bar chart in the main text), which compares Λ for the six highest-risk cities across the three model tiers for each disease. Two summary metrics are used in the analysis:

- $P(\geq 1)$: probability of at least one importation (Equation 1). Useful as a communication metric but saturates quickly toward 1 for high-risk cities.
- Λ : expected number of imported cases (the raw Poisson intensity). Remains on a linear scale even when $P(\geq 1) \rightarrow 1$ and allows direct quantitative comparison across cities and models.

Country-level importation contributions

The additive structure of the Poisson model allows city-level importation risk to be attributed to individual source countries. Total importation intensity decomposes as:

$$\Lambda_{v,d} = \sum_c \lambda_{c,v,d} \quad (20)$$

where $\lambda_{c,v,d} = N_{c,v} \cdot I_{c,d} \cdot p_d$ is the contribution from source country c to destination city v for disease d . Using the schedule-driven model (which separates WC-fan and background streams), three summaries are produced:

1. **Country importation ranking:** the top ten source countries by total expected importations (summed across all destination cities, $\sum_v \lambda_{c,v,d}$) are shown for each disease with 95% Monte Carlo uncertainty intervals (Figure ?? in the main text).
2. **Travel-stream decomposition:** for the top 20 countries by importation burden, the fraction of $\lambda_{c,v,d}$ attributable to the WC-fan stream versus the background stream is shown separately for each disease (Figure 7). Countries with WC teams are expected to show a non-zero WC-fan component; non-qualified countries will show only background travel contributions.
3. **Country $\times$ city importation heatmap:** a matrix visualisation of $\lambda_{c,v,d}$ summed over both streams, with rows ordered by total importation burden and columns by destination city (Appendix Figures 8–12). This reveals the spatial concentration of

344 risk: WC-qualified countries show importation concentrated in their specific match
345 venues, while non-qualified countries show diffuse contributions spread via T-100 rout-
346 ing fractions.

347 Disease data sources

348 Dengue

349 Monthly country-level dengue case counts were obtained from the WHO Global Dengue
350 Surveillance dataset [7]. Dengue is the primary focus of the WC-adjusted and schedule-
351 driven models because it is highly seasonal (peaking in June in the Southern Hemisphere,
352 where several WC-qualified nations are located), is endemic across much of Latin America,
353 the Caribbean, and Southeast Asia, and has a substantial fraction of infections that are
354 asymptomatic or mild enough to allow continued travel. For all three models, the mean June
355 case count per country over 2024–2025 was used directly, divided by national population to
356 obtain a per-capita incidence. Countries with no data were excluded.

357 Malaria

358 Country-level malaria incidence rates for 2024 (cases per 1,000 population) were obtained
359 from the WHO Global Malaria Programme National Unit Data [8]. Annual incidence was
360 divided by 12 to obtain an approximate monthly rate. Malaria was included because several
361 WC-qualified countries (particularly in sub-Saharan Africa and parts of Asia) have high
362 endemic transmission, and even low traveller infection probabilities translate into meaningful
363 importation risk given the large travel volumes from Latin America and Africa. Countries
364 were recoded to match naming conventions in the travel datasets where necessary (e.g.,
365 “Democratic Republic of the Congo” → “Zaire (formerly DRC)”).

366 Measles and pertussis

367 Annual country-level incidence rates (cases per 1,000,000 population) for measles and per-
368 tussis were obtained from the WHO Immunization Data portal [9]. Annual incidence was
369 divided by 12 to obtain an approximate monthly rate. Measles was included because of its
370 resurgence in multiple regions due to declining vaccination coverage, and because it has a
371 high infectiousness ($R_0 \approx 12$ –18) that makes even a single importation event meaningful.
372 Pertussis was included because its catarrhal phase closely resembles a common cold, making
373 it likely that infected individuals continue to travel internationally; its p_d value (Table 2) is
374 accordingly the highest of the five diseases.

Influenza

Weekly country-level positive specimen counts were obtained from the WHO FluNet / Global Influenza Surveillance and Response System (GISRS) [9]. FluNet reports laboratory-confirmed influenza A and B specimens submitted by national sentinel surveillance sites; the combined total (INF_ALL) is used here as a proxy for reported incidence, with the under-reporting correction ρ_d accounting for the gap between confirmed specimens and true community burden. June corresponds to ISO weeks 22–26. Mean June positive specimens per country were computed across 2023–2025 to produce a stable baseline signal. Country names were harmonised to COR conventions before joining (26 known mismatches recoded, e.g., “United Kingdom of Great Britain and Northern Ireland” $\rightarrow$ “United Kingdom”). Countries with zero mean June specimens were excluded.

Influenza was included because June coincides with the peak of the Southern Hemisphere winter influenza season. WC-qualified nations in South America (Brazil, Argentina, Uruguay, Colombia, Ecuador, Peru, Bolivia, Paraguay, Chile, Venezuela) and Oceania (Australia, New Zealand) carry substantial seasonal influenza burden during the tournament window, making this the first importation risk analysis for the 2026 World Cup to quantify influenza alongside vector-borne and vaccine-preventable diseases.

Model parameters

Table 2 summarises the disease-specific parameters used in all three models. The adjustment factor ρ_d captures the combined effect of under-reporting in routine surveillance and the fraction of reported cases that represent actively infectious individuals at a given time. Values were assigned based on published burden studies: dengue and pertussis are the most severely under-reported diseases, with global detection fractions well below 20% in most endemic settings [10–12]; malaria detection at $\rho_d = 0.20$ is consistent with WHO World Malaria Report methodology [8]; and measles at $\rho_d = 0.60$ reflects the higher reporting rates typical of middle- and high-income source countries with mandatory notification [13]. The travel infectiousness probability p_d reflects the likelihood that a person with the disease travels internationally while infectious, and was informed by the typical severity and duration of symptoms relative to the incubation period and infectious window.

Table 2: Disease-specific model parameters. The ρ_d literature range is derived from published burden and under-reporting studies; the adopted value falls within or at the conservative end of that range. All p_d values are supported by travel medicine literature. Key references are given for ρ_d ; p_d values are based on biological plausibility and importation-model precedent.

Disease	ρ_d range	ρ_d	p_d	Rationale for p_d	Key ref. (ρ_d)
Dengue	0.06–0.26	0.10	0.50	Mild viraemic phase; ~40% viremic on arrival	[10, 11]
Influenza	0.033–0.20	0.10	0.50	Moderate illness; many travel during early illness; SH winter peak in June	[9]
Malaria	0.10–0.35	0.20	0.30	Severe illness; most febrile after return	[8]
Measles	0.40–0.80	0.60	0.05	Prostrating; ambulatory only in short pre-rash window	[13]
Pertussis	0.01–0.10	0.10	0.70	Catarrhal stage mimics common cold; unrecognised	[12]

Table 3: NTTO 2026 international visitor projections and derived growth factors used in the WC-adjusted model. Source: US National Travel and Tourism Office Forecast Report (March 2025).

Country	2024 (000s)	2026 (000s)	ϕ_c	WC team?
Canada	20,241	23,015	1.137	Yes (host)
Mexico	16,990	19,557	1.151	Yes (host)
United Kingdom	4,037	4,467	1.107	Yes (England, Scotland)
Japan	1,844	2,600	1.410	Yes
China	1,626	2,641	1.624	No
Germany	1,995	2,223	1.114	Yes
Brazil	1,910	2,341	1.226	Yes
South Korea	1,700	2,018	1.187	Yes
India	2,190	2,541	1.161	No
France	1,706	1,923	1.127	Yes
Australia	1,025	1,244	1.214	Yes
Italy	1,124	1,314	1.169	No
Total international	72,390	85,017	1.174	

NTTO 2026 visitor projections for top 12 source markets

Software

All analyses were performed in R (version 4.5.2). The main importation pipeline (`importationRisk_main_with_uncertainty.R`) uses `tidyverse` for data wrangling and vi-

sualisation, `readxl` for Excel data ingestion, `janitor` for column-name standardisation, `lubridate` for date operations, and `cowplot` for multi-panel figure composition. Monte Carlo uncertainty propagation (5,000 Uniform draws on ρ_d and p_d) is integrated directly into the main pipeline via the vectorised `compute_mc_summary()` helper. A dedicated data preparation script (`t100_routing_prep.R`) processes BTS T-100 data and must be run once before the main pipeline to generate `Data/t100_routing_fractions.csv`. FluNet influenza data are downloaded automatically from the WHO FluNet API on first run and cached locally. Code and data are available at:

https://github.com/jdiestra1977/FIFA_worldCup_2026_risk

Appendix B: Supplementary Results

Parameter uncertainty direction

Figure 2 shows the directional asymmetry of Monte Carlo uncertainty for each disease across all venue cities. The axes compare the upside ratio ($\Lambda_{hi}/\Lambda_{median}$) against the downside ratio ($\Lambda_{median}/\Lambda_{lo}$). Points that fall above the dashed diagonal have CIs that extend further above the median than below it—these diseases may be underestimated at the central parameter values. Points below the diagonal carry CIs that compress below the median, indicating conservative upper bounds.

Dengue and influenza consistently fall above the diagonal because their central detection rates sit near the lower end of the literature range. Pertussis sits well below the diagonal: the central $\rho = 0.10$ is the *upper* bound of its range, so the plausible space is almost entirely downward. Malaria and measles occupy intermediate, roughly symmetric positions.

Importation risk overview: Baseline and WC-adjusted models

Figures 3 and 4 show the same $P(\geq 1)$ heatmap as Figure ?? in the main text, but for the Baseline (Model 1) and WC-adjusted (Model 2) tiers respectively. City and disease ordering are identical to Figure ?? to allow direct visual comparison across all three model tiers. Dengue importation probability is near-certain (> 0.99) at the highest-risk cities under all three models, confirming that this finding is not an artefact of the schedule-driven travel assumptions. The step from Model 1 to Model 2 raises probabilities modestly for most disease–city pairs; the step to Model 3 (main text) introduces the schedule-based concentration at specific match venues.

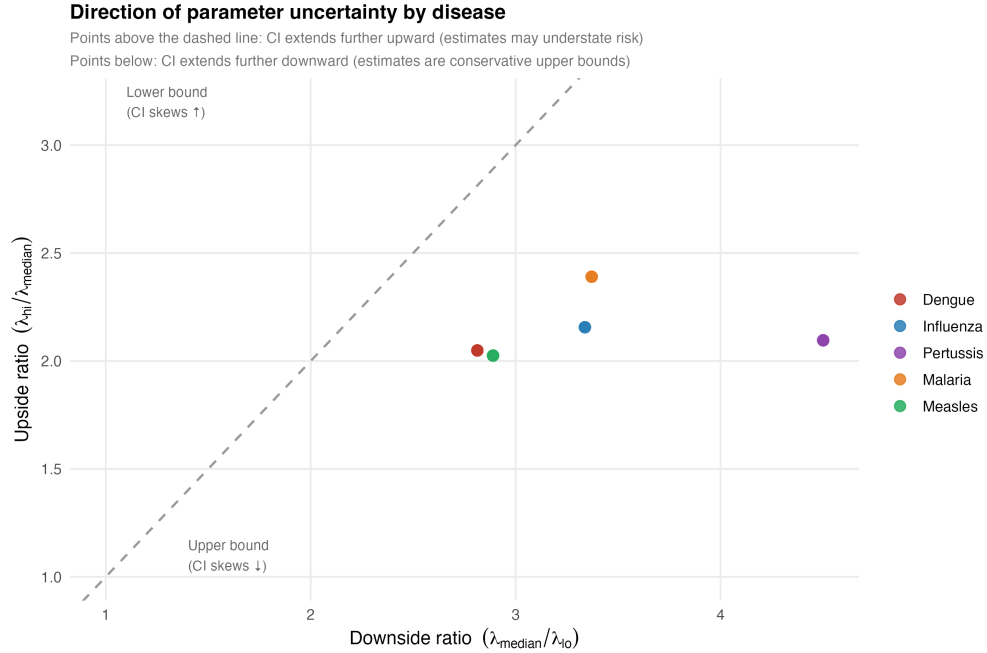


Figure 2: Direction of Monte Carlo parameter uncertainty by disease. Each point represents one disease–city pair under the schedule-driven model. Upside ratio = $\Lambda_{hi}/\Lambda_{median}$; downside ratio = $\Lambda_{median}/\Lambda_{lo}$. Points above the dashed diagonal: CI skews upward (central estimate near lower bound of plausible range). Points below: CI skews downward (central estimate near upper bound). Dengue and influenza may be underestimated; pertussis estimates are conservative upper bounds.

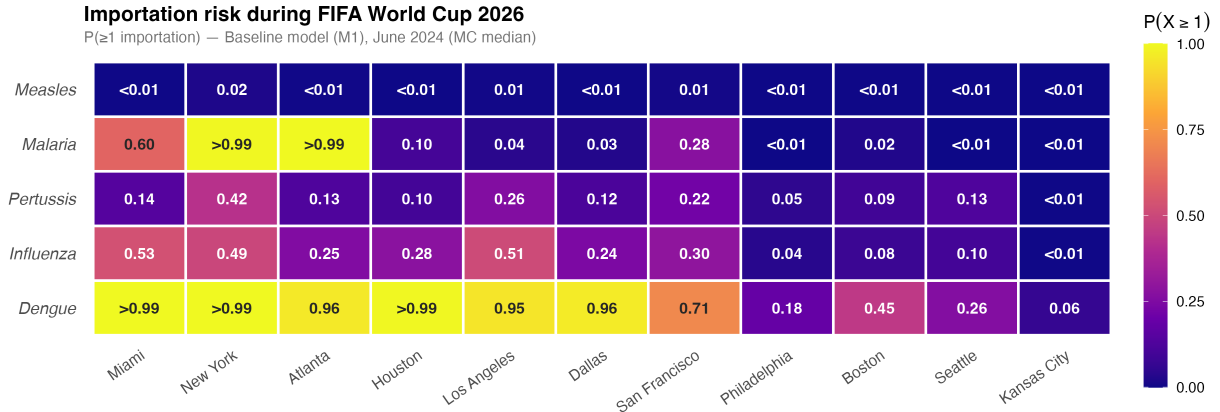


Figure 3: Importation risk overview under the Baseline model (Model 1): $P(\geq 1 \text{ importation})$ using routine June 2024 COR arrivals with no World Cup growth adjustment (MC median, 5,000 draws). Same city and disease ordering as Figure ??.

Three-model comparison of importation intensity

Figure 5 compares expected importation intensity (Λ , median and 95% MC uncertainty interval) across the three nested model tiers for the six highest-risk US cities. The near-

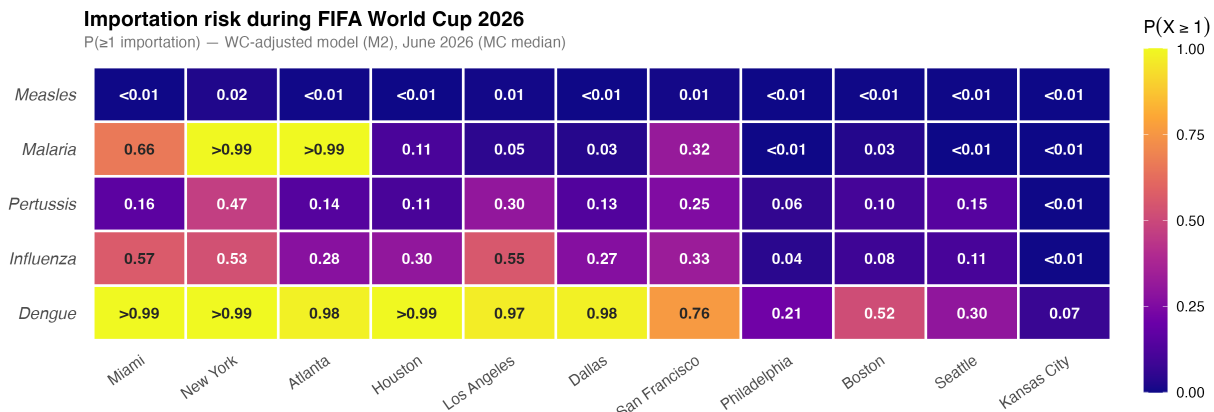


Figure 4: Importation risk overview under the WC-adjusted model (Model 2): $P(\geq 1 \text{ importation})$ using June 2026 projected arrivals scaled by NTTD growth factors (MC median, 5,000 draws). Same city and disease ordering as Figure ??.

uniform step-up from Model 1 to Model 2 reflects proportional amplification driven by the WC growth factor; the further step to Model 3 introduces modest city-specific divergence through schedule-based fan routing, without changing city or disease rankings.

Dengue stream decomposition by source country

Figure 6 shows the dengue-specific travel stream decomposition for the top 20 source countries: background travel (blue) vs. WC fans (orange). The full five-disease stream panel is Figure 7.

WC-fan vs. background travel stream decomposition

Figure 7 decomposes the top 20 contributing countries by travel stream (WC fans vs. background) for each of the five diseases. For WC-qualified countries, the WC-fan stream contributes a meaningful but typically minority share of the total importation intensity, consistent with the $\approx 8.3\%$ fan fraction derived from the growth factor decomposition. Countries with the largest WC fan contributions are those that (1) have a qualified team, (2) have high disease incidence, and (3) have their team’s matches concentrated at a small number of venue cities, increasing the local fan density at those venues.

Non-qualified countries, regardless of their disease burden, contribute exclusively through the background stream; their presence in the top 20 reflects high June travel volumes to the United States rather than any WC-specific effect. This decomposition highlights that the background stream dominates total importation risk for most source countries: the

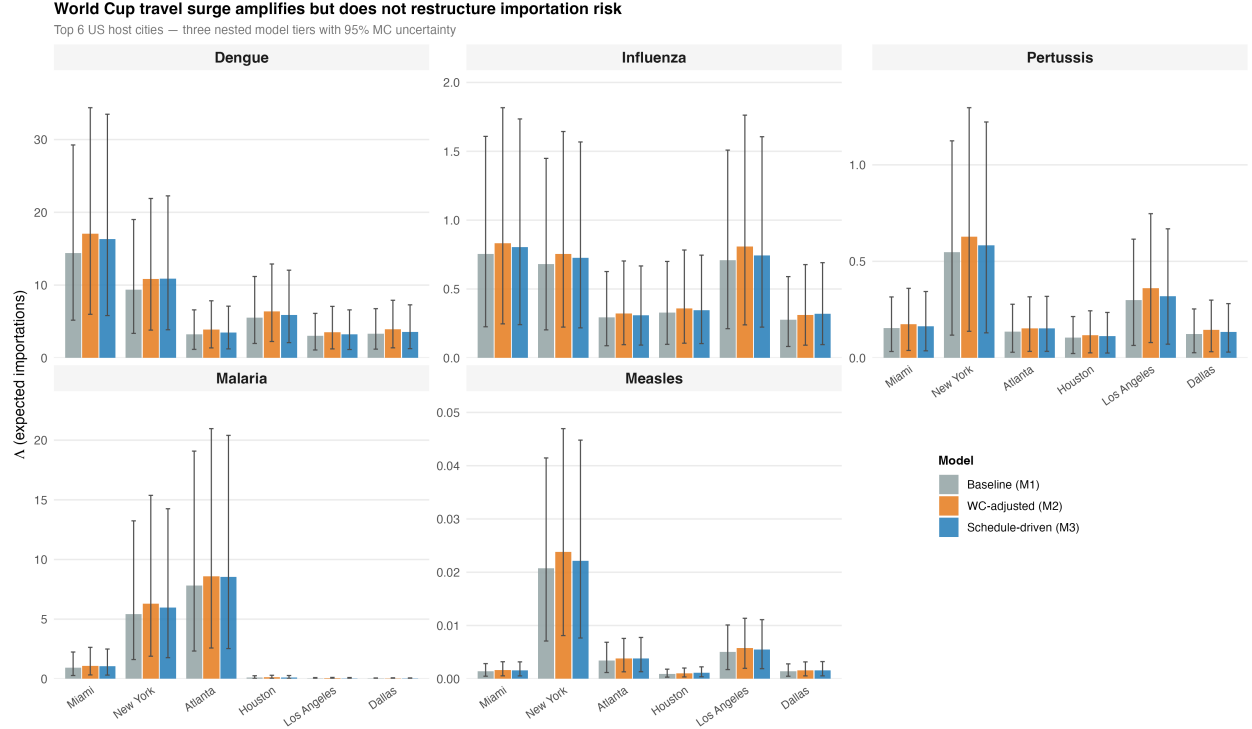


Figure 5: Three-model comparison of expected importations (Λ) for the six highest-risk US venue cities. Each panel shows one disease; bars are grouped by city and filled by model tier: Baseline (Model 1), WC-adjusted (Model 2), and Schedule-driven (Model 3). Bars show the MC median; whiskers span the 95% uncertainty interval (5,000 Uniform draws on ρ_d and p_d). The WC surge amplifies risk proportionally; schedule-based routing introduces modest city-specific divergence but leaves rankings unchanged.

WC tournament amplifies risk at specific cities and for specific diseases, but baseline travel patterns set the underlying risk floor across all cities and diseases.

Country $\times$ city importation heatmaps

Figures 8–12 present the country $\times$ city importation matrix ($\lambda_{c,v,d}$) for each disease. The heatmaps reveal a consistent spatial pattern: WC-qualified countries from high-incidence regions show importation concentrated at the venue cities where their team plays, producing a block-diagonal structure aligned with the match schedule. Non-qualified countries show importation spread more diffusely across cities in proportion to T-100 routing fractions, reflecting their reliance on normal tourist routing patterns.

For dengue (Figure 8), the highest country–city cells correspond to Latin American WC teams (Brazil, Colombia, Ecuador, Paraguay, Uruguay, Argentina) at their respective match venues. Miami, Houston, and Los Angeles receive notable dengue contributions from multiple

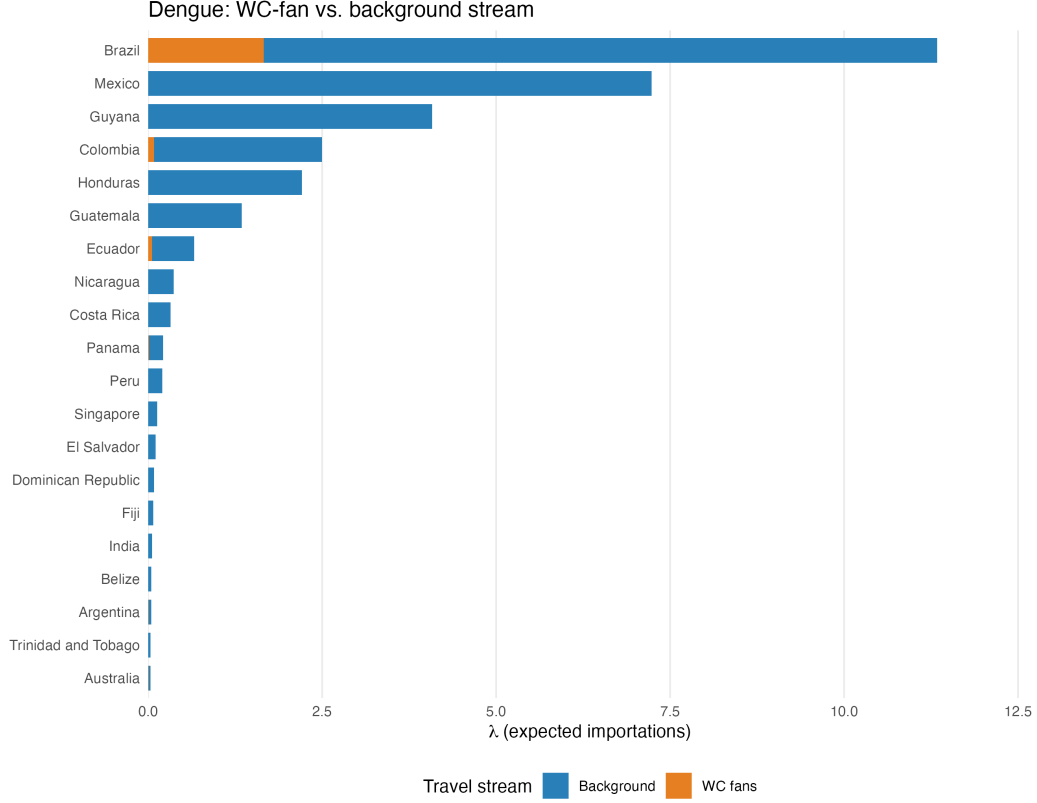


Figure 6: Dengue: importation intensity by travel stream for the top 20 source countries. Blue = background travel; orange = WC fans. Background dominates for all countries; the WC fan increment is largest for high-incidence WC-qualified nations (Brazil, Colombia, Ecuador).

Latin American and Caribbean source countries through both the WC-fan and background streams. For malaria (Figure 9), the risk matrix is sparser, with high values concentrated in West African nations at their match venues and lower diffuse contributions elsewhere.

Measles (Figure 10) shows a distinct pattern from dengue and malaria: despite having the highest ρ value, the low p_d suppresses absolute importation intensities across all country-city pairs, resulting in a more uniformly low-intensity matrix with a few notable peaks at cities hosting matches for countries with documented measles outbreaks. Pertussis (Figure 11) shows the broadest geographic spread, consistent with its global endemicity and high p_d ; the matrix includes non-trivial values from European and East Asian countries in addition to Latin American ones.

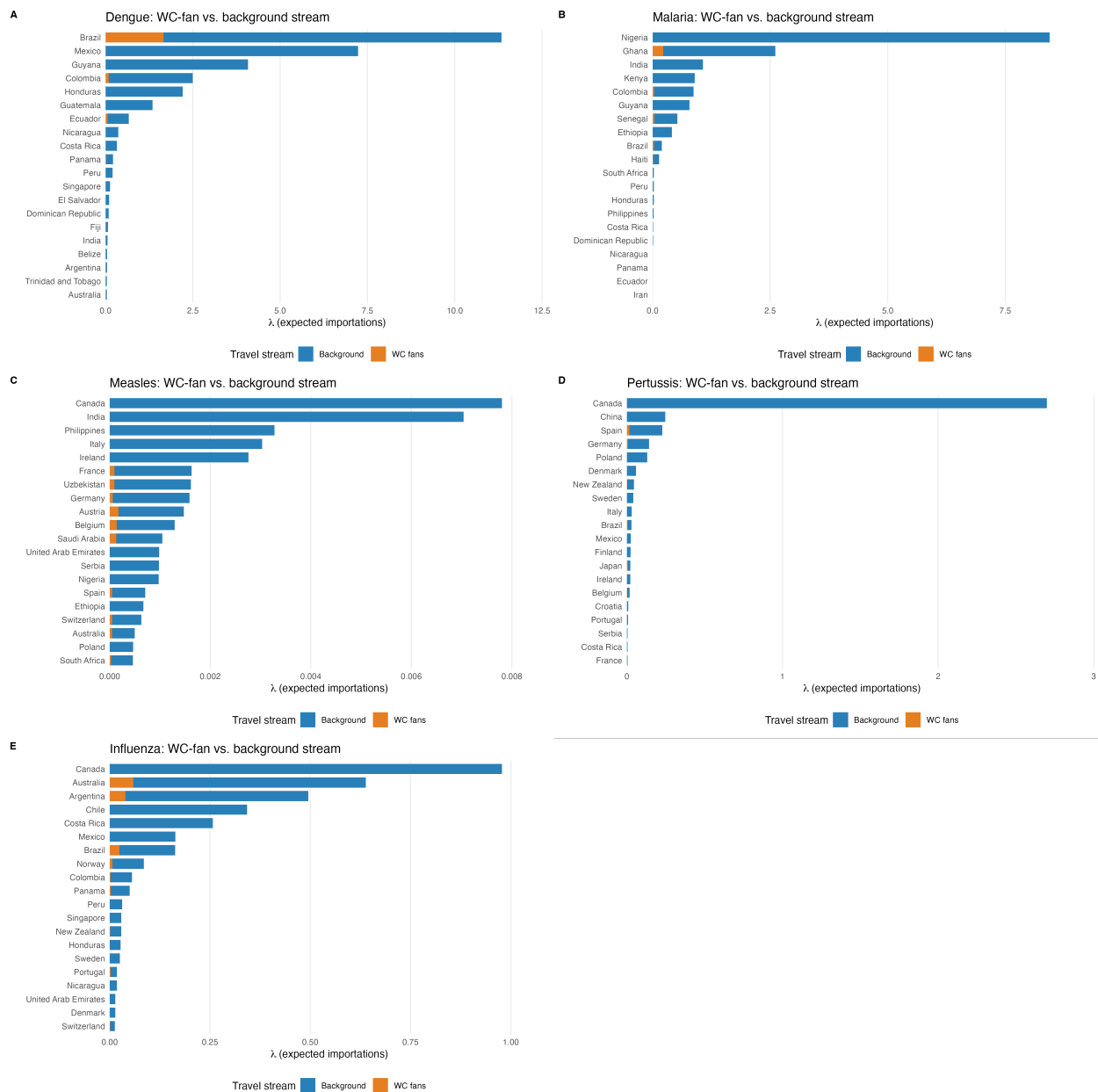


Figure 7: WC-fan vs. background travel stream decomposition for the top 20 source countries by total importation intensity, for each of the five diseases. **Orange** = WC fans; **Blue** = background. Countries with WC teams show a non-zero WC-fan component; non-qualified countries contribute only through background travel.

Appendix C: Data Limitations and Improvement Opportunities

The role of non-public data in improving importation risk estimates

The framework presented here relies entirely on publicly available data sources — a deliberate choice that ensures reproducibility and allows the model to be updated by any public

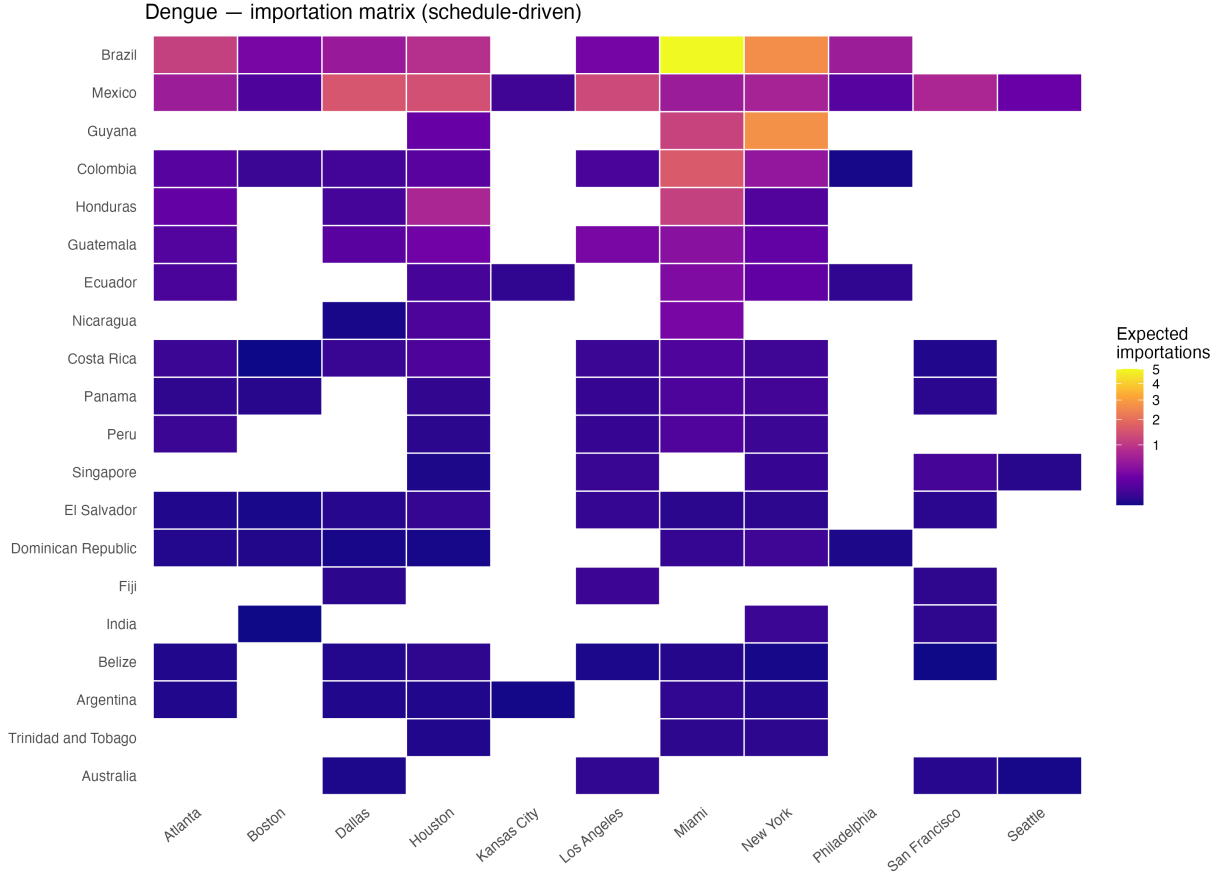


Figure 8: Dengue importation matrix: expected importations $\lambda_{c,v}$ by source country (rows, ordered by total burden) and destination city (columns) under the schedule-driven model. Plasma colour scale: bright yellow/orange = high expected importations; dark purple/blue = low.

health team with internet access. However, several categories of proprietary and restricted-access data exist that would substantially improve the precision, timeliness, and scope of the estimates, and their absence represents the primary structural limitation of the current analysis.

The most valuable single dataset for this purpose would be **airline booking and ticketing data**. Companies such as OAG (Official Aviation Guide), Amadeus Travel Intelligence, and IATA’s PaxIS programme maintain origin-to-destination itinerary-level passenger records that cover not only the nonstop international segment (as T-100 does) but the full trip, including domestic connecting legs to non-gateway cities such as Kansas City and Seattle. These records are updated in near-real time and can be filtered to the specific June–July 2026 travel window rather than relying on historical routing fractions as a proxy for future behavior. Similarly, **FIFA ticket sales records by country of purchase**, while not

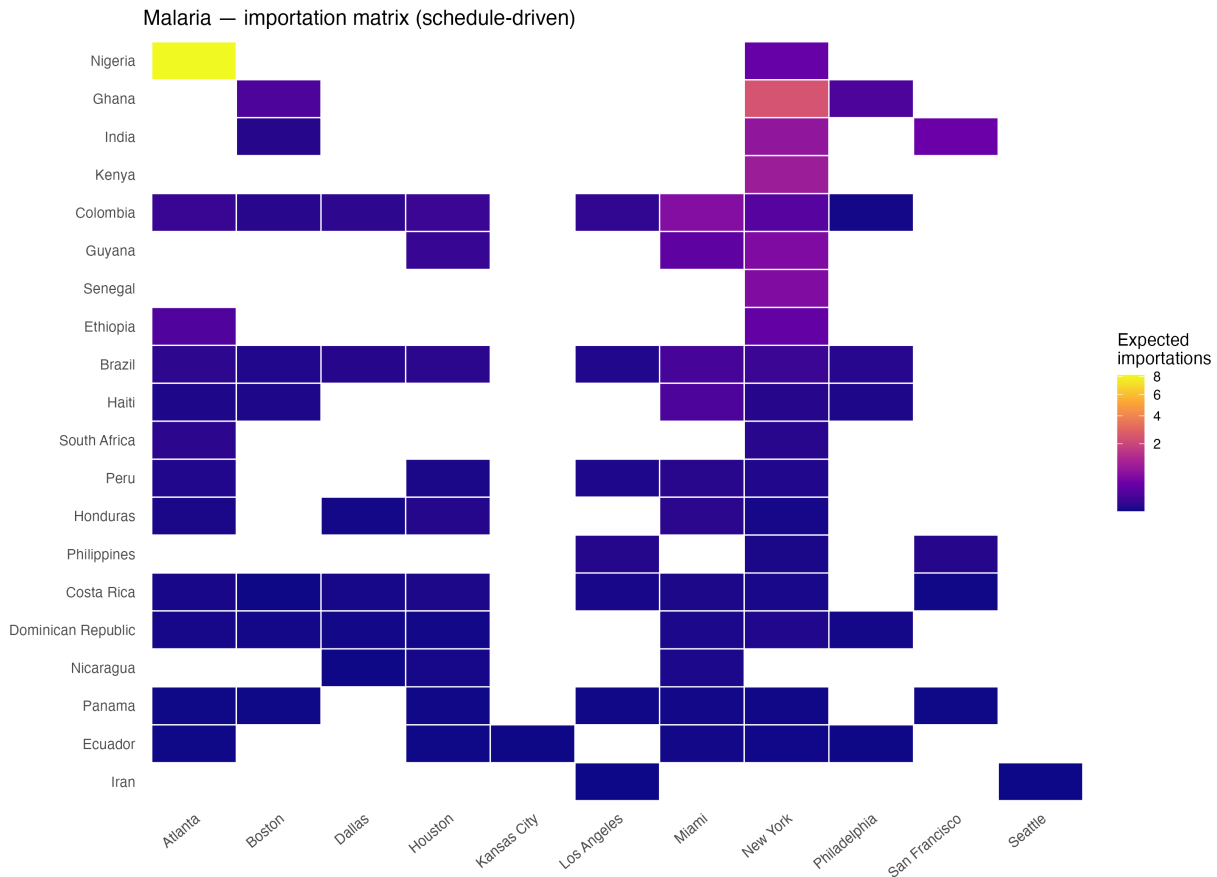


Figure 9: Malaria importation matrix. Same structure as Figure 8. Risk is concentrated in West and Central African source countries.

publicly released, would replace the entire WC-fan travel decomposition framework with a direct, event-specific measure: rather than inferring WC fans from NTTO growth factors, one could count the actual tickets sold to fans from each country and allocate them to venue cities by the printed match schedule.

Beyond travel data, the precision of disease burden estimates is constrained by the reliance on **WHO surveillance data**, which is subject to reporting lags, inconsistent national case definitions, and known under-ascertainment. National or regional health ministry **line-list or case-count data** shared through WHO regional offices, the Pan American Health Organization (PAHO), or the European Centre for Disease Prevention and Control (ECDC) often contain more current and complete figures than what appears in the publicly indexed datasets used here — particularly for dengue, where the 2025 season is not yet fully incorporated. For measles and pertussis, **age-stratified incidence and national vaccination coverage data** at the subnational level (available from national immunization programmes but not systematically published) would substantially improve the per-traveler infection

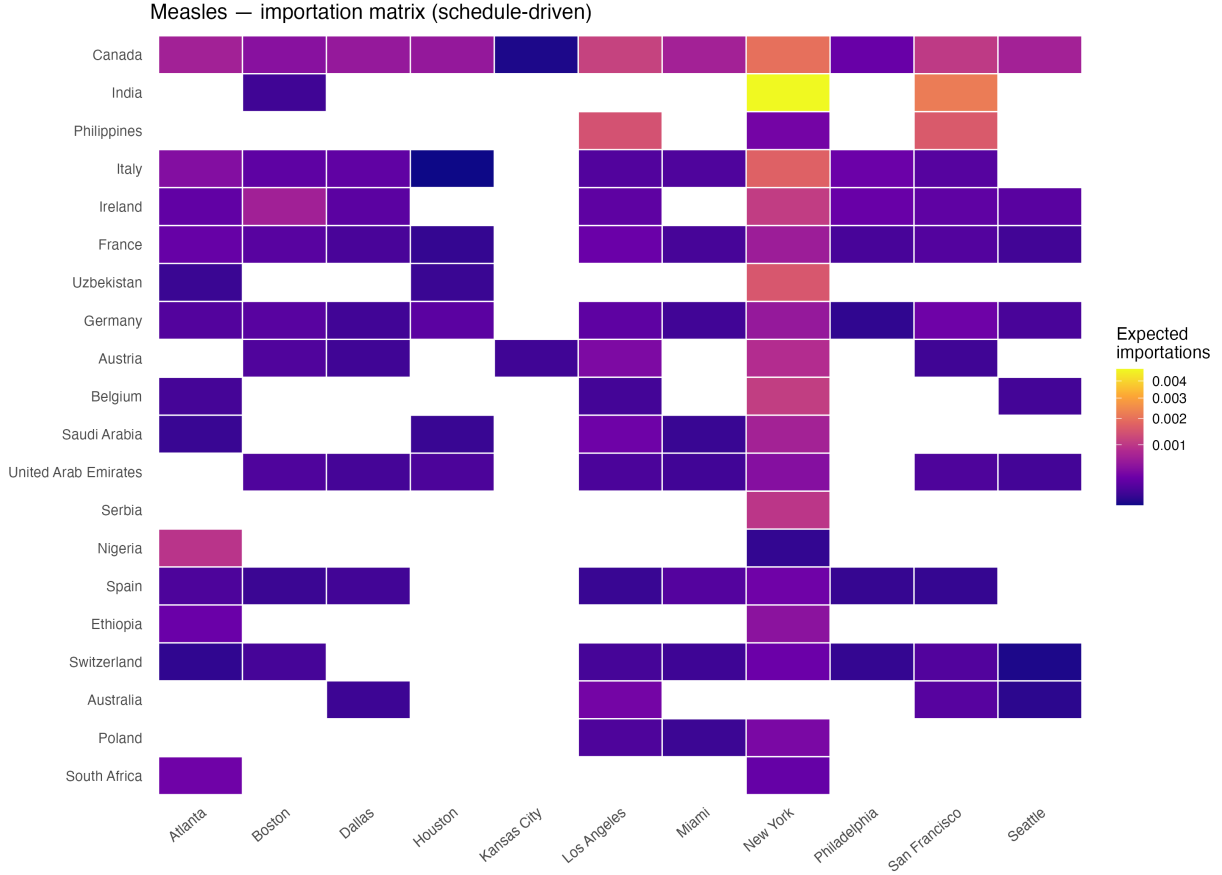


Figure 10: Measles importation matrix. Low absolute intensities across all cells reflect the low travel-while-infectious probability ($p = 0.05$).

probability $I_{c,d}$ by accounting for the fact that international travelers skew toward working-age adults, who may have markedly different immunity and infection rates than the general population.

Finally, the current model estimates importation risk but stops short of estimating local transmission risk after importation. Extending the framework in that direction would require **local *Aedes aegypti* vector surveillance data** from state and county health departments (partially available through CDC ArboNET but not consistently in machine-readable form), **hospital and public health capacity data** by city and county (available through the American Hospital Association survey but behind a licensing agreement), and **aggregated mobility data** from mobile device providers (Meta, Google) to calibrate the spatial spread of fans within and between cities. None of these datasets are currently accessible through open public APIs at the geographic resolution needed for this analysis. Establishing data-sharing agreements with the relevant agencies and commercial providers before the tournament begins — ideally through the CDC’s International Importation Re-

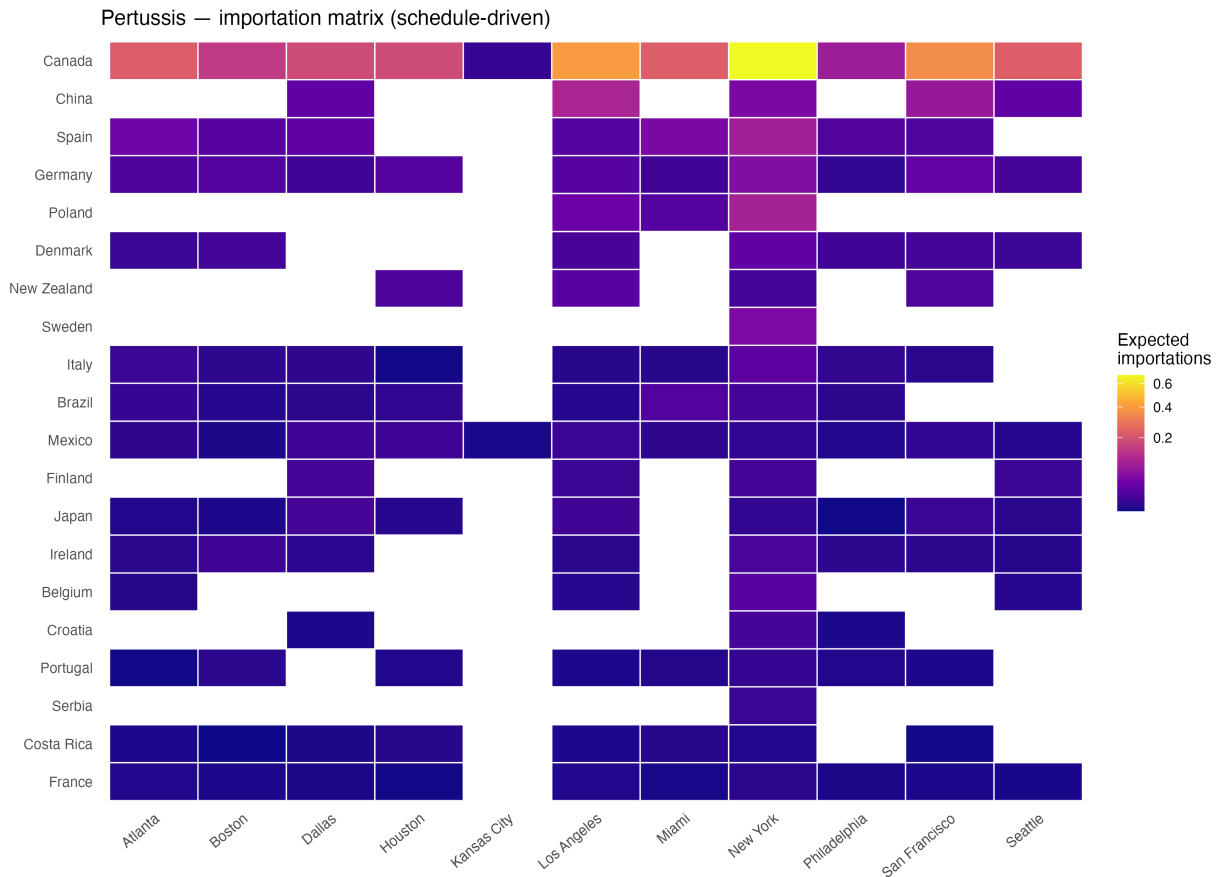


Figure 11: Pertussis importation matrix. The broad geographic spread reflects pertussis’s global endemicity and high travel-while-infectious probability ($p = 0.70$).

sponse Branch or the Council of State and Territorial Epidemiologists (CSTE) — would
allow these estimates to be refined and operationalized in real time during June–July 2026.

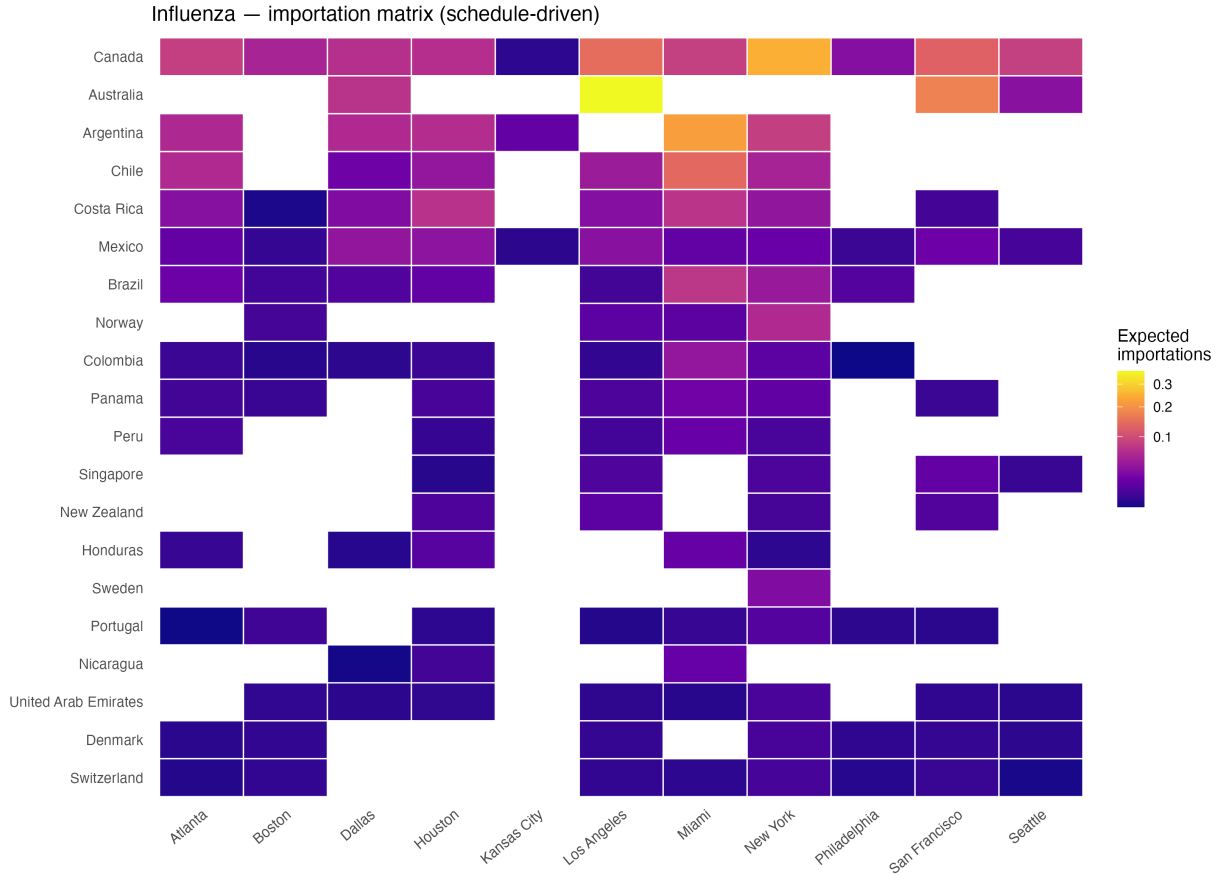


Figure 12: Influenza importation matrix: expected importations $\lambda_{c,v}$ by source country (rows, ordered by total burden) and destination city (columns) under the schedule-driven model. Plasma colour scale: bright yellow/orange = high; dark purple/blue = low. Risk is highest for Southern Hemisphere nations (Brazil, Argentina, Colombia) whose June peak transmission coincides with the tournament window. Background travel from high-volume non-qualified nations (e.g., Canada, Mexico, United Kingdom) contributes a broad diffuse signal across all cities.

Appendix D: Qualified Nations — 2026 FIFA World Cup

Country	Confederation	Host nation
Japan	AFC	No
Iran	AFC	No
South Korea	AFC	No

Continued on next page

Country	Confederation	Host nation
Australia	AFC	No
Qatar	AFC	No
Saudi Arabia	AFC	No
Uzbekistan	AFC	No
Jordan	AFC	No
Iraq	AFC	No
Morocco	CAF	No
Tunisia	CAF	No
Egypt	CAF	No
Algeria	CAF	No
Ghana	CAF	No
Cape Verde	CAF	No
Senegal	CAF	No
South Africa	CAF	No
Ivory Coast	CAF	No
DR Congo	CAF	No
Canada	CONCACAF	Yes
Mexico	CONCACAF	Yes
United States	CONCACAF	Yes
Panama	CONCACAF	No
Curaçao	CONCACAF	No
Haiti	CONCACAF	No
Argentina	CONMEBOL	No
Brazil	CONMEBOL	No
Ecuador	CONMEBOL	No
Paraguay	CONMEBOL	No
Uruguay	CONMEBOL	No
Colombia	CONMEBOL	No
England	UEFA	No
France	UEFA	No
Croatia	UEFA	No
Portugal	UEFA	No
Norway	UEFA	No

Continued on next page

Country	Confederation	Host nation
Germany	UEFA	No
Netherlands	UEFA	No
Switzerland	UEFA	No
Scotland	UEFA	No
Spain	UEFA	No
Austria	UEFA	No
Belgium	UEFA	No
Bosnia and Herzegovina	UEFA	No
Sweden	UEFA	No
Turkey	UEFA	No
Czech Republic	UEFA	No
New Zealand	OFC	No

Note: Confederation abbreviations — AFC: Asian Football Confederation; CAF: Confederation of African Football; CONCACAF: Confederation of North, Central America and Caribbean Association Football; CONMEBOL: South American Football Confederation; UEFA: Union of European Football Associations; OFC: Oceania Football Confederation. Source: https://en.wikipedia.org/wiki/2026_FIFA_World_Cup_qualification (accessed April 2026).

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